

Variational Inference

Problem

Assume n observations $\mathbf{x}$ and m latent variables $\mathbf{z}$. Bayes rule states that

$$\underbrace{p(\mathbf{z}|\mathbf{x})}_{\text{Posterior}} = \underbrace{p(\mathbf{z})}_{\text{Prior}} \cdot \frac{\overbrace{p(\mathbf{x}|\mathbf{z})}^{\text{Likelihood}}}{\underbrace{\int p(\mathbf{x}|\mathbf{z})p(\mathbf{z}) \, \mathrm{d}\mathbf{z}}_{\text{Marginal likelihood}}}.$$

We want to evaluate the posterior $p(\mathbf{z}|\mathbf{x})$. However, the marginal likelihood $p(\mathbf{x})$ is often hard to compute.

Applications mentioned in Blei et al. (2017)

- Computational biology
- Computer vision and robotics
- Computational neuroscience
- Natural language processing and speech recognition
- Other applications (notably generative models)

Variational inference

Variational inference (VI) attempts to *reframe* the Bayesian inference problem as an optimization problem. Optimization, which involves taking derivatives instead of integrating, is generally faster (and easier?).

Variational objective

Find a density $q^*(z)$ from a family of densities $\mathcal{D}$ that approximates the posterior

$$q^*(z) = \operatorname{argmin}_{q(z) \in \mathcal{D}} \operatorname{KL}(q(z) \| p(z|x)),$$

where the Kullback-Leibler divergence is defined as

$$\operatorname{KL}(q(z) \| p(z|x)) = \int q(z) \log \frac{q(z)}{p(z|x)} \mathrm{d}z.$$

Evidence lower bound (ELBO)

Writing out the Kullback-Leibler divergence, we find

$$\begin{aligned}\text{KL}(q(\mathbf{z})\|p(\mathbf{z}|\mathbf{x})) &= \mathbb{E}_{q(\mathbf{z})}[\log q(\mathbf{z})] - \mathbb{E}_{q(\mathbf{z})}[\log p(\mathbf{z}|\mathbf{x})] \\ &= \mathbb{E}_{q(\mathbf{z})}[\log q(\mathbf{z})] - \mathbb{E}_{q(\mathbf{z})}[\log p(\mathbf{z}, \mathbf{x})] + \mathbb{E}_{q(\mathbf{z})}[\log p(\mathbf{x})] \\ &= \mathbb{E}_{q(\mathbf{z})}[\log q(\mathbf{z})] - \mathbb{E}_{q(\mathbf{z})}[\log p(\mathbf{z}, \mathbf{x})] + \underbrace{\log p(\mathbf{x})}_{\text{Unknown}}\end{aligned}$$

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Therefore instead maximize the ELBO, defined as

$$\begin{aligned}\text{ELBO}(q) &= -(\text{KL}(q(\mathbf{z})\|p(\mathbf{z}|\mathbf{x})) - \log p(\mathbf{x})) \\ &= \mathbb{E}_{q(\mathbf{z})}[\log p(\mathbf{z}, \mathbf{x})] - \mathbb{E}_{q(\mathbf{z})}[\log q(\mathbf{z})].\end{aligned}$$

ELBO intuition

$$\begin{aligned}\text{ELBO}(q) &= \mathbb{E}_{q(\mathbf{z})}[\log p(\mathbf{x}|\mathbf{z})] + \mathbb{E}_{q(\mathbf{z})}[\log p(\mathbf{z})] - \mathbb{E}_{q(\mathbf{z})}[\log q(\mathbf{z})] \\ &= \mathbb{E}_{q(\mathbf{z})}[\log p(\mathbf{x}|\mathbf{z})] - \mathbb{E}_{q(\mathbf{z})} \left[\log \frac{q(\mathbf{z})}{p(\mathbf{z})} \right] \\ &= \underbrace{\mathbb{E}_{q(\mathbf{z})}[\log p(\mathbf{x}|\mathbf{z})]}_{\text{q fits data}} - \underbrace{\text{KL}(q(\mathbf{z})\|p(\mathbf{z}))}_{\text{q satisfies prior}}\end{aligned}$$

ELBO intuition

$$\begin{aligned}\text{ELBO}(q) &= \mathbb{E}_{q(\mathbf{z})}[\log p(\mathbf{x}|\mathbf{z})] + \mathbb{E}_{q(\mathbf{z})}[\log p(\mathbf{z})] - \mathbb{E}_{q(\mathbf{z})}[\log q(\mathbf{z})] \\ &= \mathbb{E}_{q(\mathbf{z})}[\log p(\mathbf{x}|\mathbf{z})] - \mathbb{E}_{q(\mathbf{z})} \left[\log \frac{q(\mathbf{z})}{p(\mathbf{z})} \right] \\ &= \underbrace{\mathbb{E}_{q(\mathbf{z})}[\log p(\mathbf{x}|\mathbf{z})]}_{\text{q fits data}} - \underbrace{\text{KL}(q(\mathbf{z})\|p(\mathbf{z}))}_{\text{q satisfies prior}}\end{aligned}$$

Additionally, since

$$\log p(\mathbf{x}) = \text{ELBO}(q) + \text{KL}(q(\mathbf{z})\|p(\mathbf{z}|\mathbf{x})) \geq \text{ELBO}(q),$$

it gives a lower bound for the log marginal likelihood.

To make optimization easier, we have to constrain the family of densities $\mathcal{D}$.

Mean-field variational family

Assume latent variables z_j are mutually independent, and each governed by their own factor in the variational density

$$q(\mathbf{z}) = \prod_{j=1}^m q_j(z_j).$$

Note, this family cannot model correlations in the posterior distribution. Also see that we do not make any parametric assumptions about the individual $q_j(z_j)$, as this is instead derived for every particular problem.

Applying mean-field assumption to ELBO

$$\begin{aligned}\text{ELBO}(q) &= \mathbb{E}_{q(\mathbf{z})}[\log p(\mathbf{z}, \mathbf{x})] - \mathbb{E}_{q(\mathbf{z})}[\log q(\mathbf{z})] \\ &= \int \prod_{i=1}^m q_i(z_i) \log p(\mathbf{z}, \mathbf{x}) \, d\mathbf{z} - \int \prod_{i=1}^m q_i(z_i) \log \prod_{i=1}^m q_i(z_i) \, d\mathbf{z}.\end{aligned}$$

Optimizing ELBO

Optimize ELBO with respect to a single variational density $q_j(z_j)$ and assume others are fixed

$$\begin{aligned}\text{ELBO}(q_j) &= \int \prod_{i=1}^m q_i(z_i) \log p(\mathbf{z}, \mathbf{x}) \, d\mathbf{z} - \int \prod_{i=1}^m q_i(z_i) \log \prod_{i=1}^m q_i(z_i) \, d\mathbf{z} \\ &\propto \int \prod_{i=1}^m q_i(z_i) \log p(\mathbf{z}, \mathbf{x}) \, d\mathbf{z} - \int q_j(z_j) \log q_j(z_j) \, dz_j \\ &= \int q_j(z_j) \left(\int \prod_{i \neq j} q_i(z_i) \log p(\mathbf{z}, \mathbf{x}) \, d\mathbf{z}_{-j} \right) dz_j - \int q_j(z_j) \log q_j(z_j) \, dz_j \\ &= \int q_j(z_j) \mathbb{E}_{q(\mathbf{z}_{-j})} [\log p(\mathbf{z}, \mathbf{x})] \, dz_j - \int q_j(z_j) \log q_j(z_j) \, dz_j\end{aligned}$$

Optimizing ELBO

To derive the optimal $q_j^*(z_j)$ we follow Bishop (2006) and define

$$\log \tilde{p}(\boldsymbol{x}, z_j) \propto \mathbb{E}_{q(\boldsymbol{z}_{-j})} [\log p(\boldsymbol{z}, \boldsymbol{x})].$$

Optimizing ELBO

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$$\log \tilde{p}(\mathbf{x}, z_j) \propto \mathbb{E}_{q(z_{-j})} [\log p(\mathbf{z}, \mathbf{x})].$$

Then

$$\begin{aligned} \text{ELBO}(q_j) &\propto \int q_j(z_j) \log \tilde{p}(\mathbf{x}, z_j) \, dz_j - \int q_j(z_j) \log q_j(z_j) \, dz_j \\ &= - \int q_j(z_j) \log \frac{q_j(z_j)}{\tilde{p}(\mathbf{x}, z_j)} \, dz_j = - \text{KL}(q_j(z_j) \| \tilde{p}(\mathbf{x}, z_j)) \end{aligned}$$

Optimizing ELBO

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Then

$$\begin{aligned} \text{ELBO}(q_j) &\propto \int q_j(z_j) \log \tilde{p}(\mathbf{x}, z_j) \, dz_j - \int q_j(z_j) \log q_j(z_j) \, dz_j \\ &= - \int q_j(z_j) \log \frac{q_j(z_j)}{\tilde{p}(\mathbf{x}, z_j)} \, dz_j = - \text{KL}(q_j(z_j) \| \tilde{p}(\mathbf{x}, z_j)) \end{aligned}$$

Maximizing ELBO with respect to q_j is equivalent to minimizing the KL divergence between $q_j(z_j)$ and $\tilde{p}(\mathbf{x}, z_j)$! This means that

$$q_j^*(z_j) \propto \exp\left(\mathbb{E}_{q(\mathbf{z}_{-j})}[\log p(\mathbf{z}, \mathbf{x})]\right).$$

Coordinate Ascent Variational Inference (CAVI)

Using

$$q_j^*(z_j) \propto \exp\left(\mathbb{E}_{q(z_{-j})}[\log p(z, \mathbf{x})]\right),$$

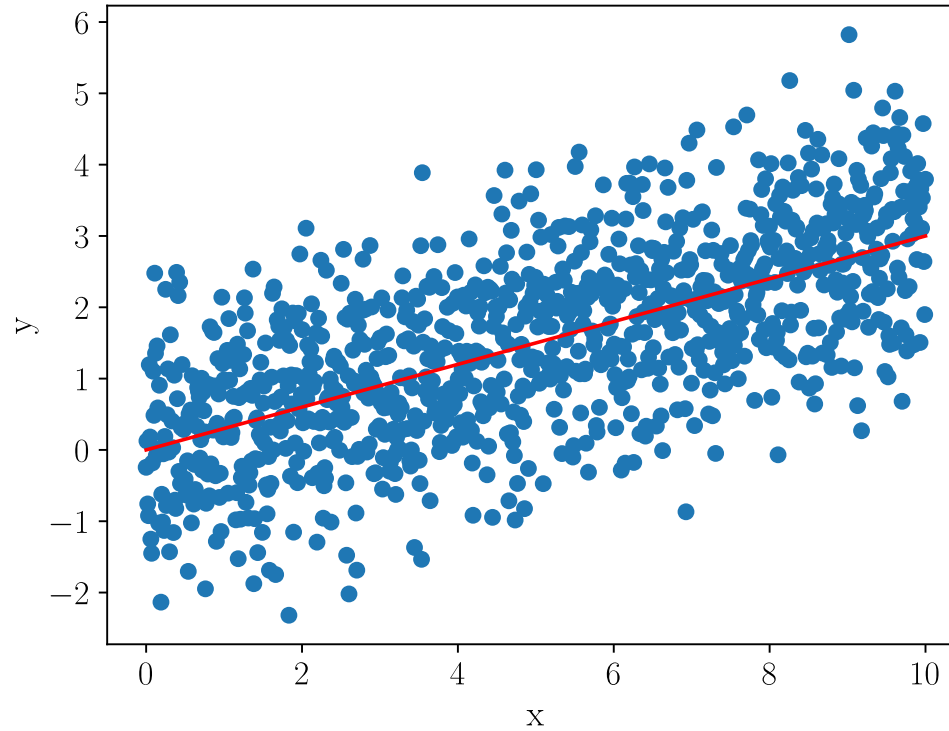
first initialize all factors $q_j(z_j)$ and then cycle through them, updating them conditional on the updates of the other.

Note that

$$p(z_j | z_{-j}, \mathbf{x}) = \frac{p(z_j, z_{-j}, \mathbf{x})}{p(z_{-j}, \mathbf{x})} \propto p(z, \mathbf{x}),$$

Example

I want to try...linear regression!



Assume the following model

- Observations $y \sim \mathcal{N}(\beta x, \sigma^2)$
- Latent variables $z = (\beta, \sigma^2)$.
- Priors
 - $\beta \sim \mathcal{N}(0, \tau^2 \sigma^2)$ (“standardized” so $\text{Var}[\beta/\sigma] = \tau$)
 - $\sigma^2 \propto \sigma^{-2}$ (improper Jeffreys prior)

Variational density for σ^2 :

Compute

$$\begin{aligned} p(\sigma^2 | \mathbf{y}, \beta) &\propto p(\mathbf{y} | \beta, \sigma^2) p(\beta) p(\sigma^2) \\ &\propto \left(\prod_{i=1}^n (\sigma^2)^{-\frac{1}{2}} \exp\left(-\frac{1}{2\sigma^2} (y_i - \beta x_i)^2\right) \right) \left((\tau^2 \sigma^2)^{-\frac{1}{2}} \exp\left(-\frac{\beta}{2\tau^2 \sigma^2}\right) \right) \left(\frac{1}{\sigma^2} \right) \\ &\propto (\sigma^2)^{-\frac{n+1}{2}-1} \exp\left(-\frac{1}{2\sigma^2} \underbrace{\left(\sum_{i=1}^n (y_i - \beta x_i)^2 + \frac{\beta^2}{\tau^2} \right)}_A \right) \end{aligned}$$

Variational density for σ^2 :

We therefore find

$$\begin{aligned} q^*(\sigma^2) &\propto \exp\left(\mathbb{E}_{q(\beta)}[\log p(\mathbf{y}, \beta, \sigma^2)]\right) \\ &\propto \exp\left(\mathbb{E}_{q(\beta)}\left[\log(\sigma^2)^{-\frac{n+1}{2}-1} - \frac{1}{2\sigma^2}A\right]\right) \\ &= (\sigma^2)^{-\frac{n+1}{2}-1} \exp\left(-\frac{1}{\sigma^2}\mathbb{E}_{q(\beta)}\left[\frac{1}{2}A\right]\right) \end{aligned}$$

Variational density for σ^2 :

We therefore find

$$\begin{aligned} q^*(\sigma^2) &\propto \exp\left(\mathbb{E}_{q(\beta)}[\log p(\mathbf{y}, \beta, \sigma^2)]\right) \\ &\propto \exp\left(\mathbb{E}_{q(\beta)}\left[\log(\sigma^2)^{-\frac{n+1}{2}-1} - \frac{1}{2\sigma^2}A\right]\right) \\ &= (\sigma^2)^{-\frac{n+1}{2}-1} \exp\left(-\frac{1}{\sigma^2}\mathbb{E}_{q(\beta)}\left[\frac{1}{2}A\right]\right) \end{aligned}$$

Recognize this as the kernel of an inverse-gamma distribution. Letting $\nu = \mathbb{E}_{q(\beta)}\left[\frac{1}{2}A\right]$ we see

$$q^*(\sigma^2) = \frac{\nu^{\frac{n+1}{2}}}{\Gamma\left(\frac{n+1}{2}\right)} (\sigma^2)^{-\frac{n+1}{2}-1} \exp\left(-\frac{\nu}{\sigma^2}\right).$$

Note this depends on β through the (so far unknown) expectation $\mathbb{E}_{q(\beta)}\left[\frac{1}{2}A\right]$.

Variational density for β :

Compute

$$\begin{aligned} p(\beta|\mathbf{y}, \sigma^2) &\propto \left(\prod_{i=1}^n (\sigma^2)^{-\frac{1}{2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - \beta x_i)^2\right) \right) \left((\tau^2 \sigma^2)^{-\frac{1}{2}} \exp\left(-\frac{\beta}{2\tau^2 \sigma^2}\right) \right) \left(\frac{1}{\sigma^2} \right) \\ &\propto \exp\left(-\frac{1}{2\sigma^2} \left(\beta^2 \left(\sum_{i=1}^n x_i^2 + \frac{1}{\tau^2} \right) - 2\beta \sum_{i=1}^n y_i x_i \right) \right) \\ &\propto \exp\left(-\frac{\sum_{i=1}^n x_i^2 + \frac{1}{\tau^2}}{2\sigma^2} \left(\beta - \frac{\sum_{i=1}^n y_i x_i}{\sum_{i=1}^n x_i^2 + \frac{1}{\tau^2}} \right)^2 \right) \end{aligned}$$

Variational density for β :

Taking expectations

$$\begin{aligned} q^*(\beta) &\propto \exp\left(\mathbb{E}_{q(\sigma^2)}[\log p(\beta|\mathbf{y}, \sigma^2)]\right) \\ &\propto \exp\left(-\frac{\sum_{i=1}^n x_i^2 + \frac{1}{\tau^2}}{2} \mathbb{E}_{q(\sigma^2)}\left[\frac{1}{\sigma^2}\right] \left(\beta - \frac{\sum_{i=1}^n y_i x_i}{\sum_{i=1}^n x_i^2 + \frac{1}{\tau^2}}\right)^2\right) \end{aligned}$$

Inserting the normalizing constants

$$q^*(\beta) = \left(2\pi \frac{\mathbb{E}_{q(\sigma^2)}\left[\frac{1}{\sigma^2}\right]^{-1}}{\underbrace{\sum_{i=1}^n x_i^2 + \frac{1}{\tau^2}}_{\sigma_\beta^2}}\right)^{-\frac{1}{2}} \exp\left(-\frac{\sum_{i=1}^n x_i^2 + \frac{1}{\tau^2}}{2} \mathbb{E}_{q(\sigma^2)}\left[\frac{1}{\sigma^2}\right] \left(\beta - \frac{\sum_{i=1}^n y_i x_i}{\underbrace{\sum_{i=1}^n x_i^2 + \frac{1}{\tau^2}}_{\mu_\beta}}\right)^2\right)$$

Computing the expectations

From $q^*(\sigma^2)$ we needed

$$\begin{aligned} 2\nu = \mathbb{E}_{q(\beta)}[A] &= \mathbb{E}_{q(\beta)} \left[\sum_{i=1}^n (y_i - \beta x_i)^2 + \frac{\beta^2}{\tau^2} \right] \\ &= \sum_{i=1}^n y_i^2 - 2 \sum_{i=1}^n y_i x_i \mathbb{E}_{q(\beta)}[\beta] + \sum_{i=1}^n x_i^2 \mathbb{E}_{q(\beta)}[\beta^2] + \frac{1}{\tau^2} \mathbb{E}_{q(\beta)}[\beta^2] \end{aligned}$$

Since $\mathbb{E}_{q(\beta)}[\beta] = \mu_\beta$ and $\mathbb{E}_{q(\beta)}[\beta^2] = \mu_\beta^2 + \sigma_\beta^2$, we find

$$\mathbb{E}_{q(\beta)}[A] = \sum_{i=1}^n y_i^2 - 2 \sum_{i=1}^n y_i x_i \mu_\beta + (\sigma_\beta^2 + \mu_\beta^2) \left(\sum_{i=1}^n x_i^2 + \frac{1}{\tau^2} \right)$$

Computing the expectations

From $q^*(\beta)$ we needed

$$\begin{aligned}
 \mathbb{E}_{q(\sigma^2)} \left[\frac{1}{\sigma^2} \right] &= \int \frac{1}{\sigma^2} \frac{\nu^{\frac{n+1}{2}}}{\Gamma\left(\frac{n+1}{2}\right)} (\sigma^2)^{-\frac{n+1}{2}-1} \exp\left(-\frac{1}{\sigma^2} \nu\right) d\sigma^2 \\
 &= \frac{\nu^{\frac{n+1}{2}}}{\Gamma\left(\frac{n+1}{2}\right)} \int (\sigma^2)^{-(\frac{n+1}{2}+1)-1} \exp\left(-\frac{1}{\sigma^2} \nu\right) d\sigma^2 \\
 &= \frac{\nu^{\frac{n+1}{2}}}{\Gamma\left(\frac{n+1}{2}\right)} \frac{\Gamma\left(\frac{n+1}{2} + 1\right)}{\nu^{\frac{n+1}{2}+1}} \\
 &= \frac{n+1}{2} \left(\frac{1}{2} \mathbb{E}_{q(\beta)}[A] \right)^{-1}
 \end{aligned}$$

Computing the ELBO

Jesus Christ...but we are not done yet! We have to iterate these updates until convergence, and this is measured by the change in ELBO.

$$\text{ELBO}(q) = \mathbb{E}_{q(\beta, \sigma^2)} [\log p(\mathbf{x}|\beta, \sigma^2)] + \underbrace{\mathbb{E}_{q(\beta, \sigma^2)} \left[\log \frac{p(\beta, \sigma^2)}{q(\beta, \sigma^2)} \right]}_{\text{still unknown}}$$

OK, we skip the details, but this is computable and we can use it to check for convergence.

Implementation and experimentation

Conditional conjugacy and stochastic variational inference (Hoffman et al., 2013)

- Nice and useful article.
- Approach for computing (approximating) the gradient ∇ELBO
 - Makes some assumptions about the model family...
- This is the way to scale the VI to super large datasets!

Automatic differentiation VI (Kucukelbir et al., 2016)

- The goal is to work for any model, and only requires that the user specifies $\log p(\boldsymbol{x}, \boldsymbol{z})$.
- Assume the variational family is a mean-field Gaussian with diagonal covariance matrix.
- Implemented in Stan (Stan Development Team, 2018)!
- Some tests I did with the multivariate regression model
 - Relatively good approximation, but not perfect.
 - Faster than MCMC, but much slower than CAVI (my implementation was NOT super great...)
 - Much easier to implement!

Conclusion

Open problems and discussion

- Other distance measures than KL
- Alternatives to mean-field
 - Add dependencies between latent variables
- Statistical properties of VI
- “Interface between variational inference and MCMC” (Blei et al., 2017)
- Relation to Variational Autoencoders (Kingma and Welling, 2022)

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